UCLA Math 131A, Winter 2025
Midterm 1, Lecture 4

Instructor: Linfeng Li

Monday, January 27

Please print your name and student ID below.

Name: ______________________________

Student ID: _________________________

Please read all of the following rules before proceeding.

- The exam worth 40 points in total.
- Show all your work, and precisely state any theorems that you use.
- You are not allowed to use calculators or notes during the test, but you may use one-page, double-sided, and handwritten cheat sheet.
- Good luck!

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
Total:	40	

1. (10 points) Circle TRUE or FALSE for each of the following statement. No justification is needed.

   (i) $\boxed{\text{True}}$ False For any $a \in \mathbb{R}$, if a is irrational then $\sqrt{a}$ is also irrational.

   (ii) True $\boxed{\text{False}}$ The set $\{r \in \mathbb{Q} : 0 \leq r \leq \sqrt{2}\}$ has both maximum and minimum.

   (iii) $\boxed{\text{True}}$ False Any finite set $S \subseteq \mathbb{R}$ has both supremum and infimum as real numbers.

   (iv) $\boxed{\text{True}}$ False For any $a, b \in \mathbb{R}$, we have $|a - b| \leq |a| + |b|$.

   (v) True $\boxed{\text{False}}$ The negation of the statement "$\forall x \in \mathbb{R}$, if x is positive, then $2x^2 > x$" is "$\exists x \in \mathbb{R}$ such that x is non-positive and $2x^2 \leq x$".

2. (10 points) Use induction to prove that for any $n \in \mathbb{N}$, we have

$$1^3 + 2^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}.$$

Solution: Let P_n be the statement $\sum_{k=1}^{n} k^3 = \frac{n^2(n+1)^2}{4}$. The base case P_1 is "$1^3 = \frac{1 \cdot 2^2}{4}$" which clearly holds. Suppose that P_n holds for some $n \in \mathbb{N}$ and we aim to prove that P_{n+1} holds. We compute

$$\begin{aligned}
\sum_{k=1}^{n+1} k^3 = \sum_{k=1}^{n} k^3 + (n+1)^3 &= \frac{n^2(n+1)^2}{4} + (n+1)^3 \\
&= \frac{(n+1)^2(n^2+4n+4)}{4} = \frac{(n+1)^2(n+2)^2}{4}.
\end{aligned}$$

Hence, P_{n+1} holds. By induction, P_n holds for any $n \in \mathbb{N}$.

3. (a) (5 points) Suppose that $S \subseteq \mathbb{R}$ is non-empty and $\sup S = m_1$ is a real number. Prove that the supremum is unique, i.e., if there exists $m_2 \in \mathbb{R}$ such that m_2 is the least upper bound of S, then $m_1 = m_2$.

   (b) (5 points) Let $x \in \mathbb{R}$. Suppose that $0 \leq x \leq \epsilon$ for any $\epsilon > 0$. Prove that $x = 0$.

Solution: (a) Let $m_2 \in \mathbb{R}$ be the least upper bound of S. Since m_1 is an upper bound of S, we get $m_2 \leq m_1$. Similarly, $m_1 \leq m_2$ since m_1 is the least upper bound of S and m_2 is an upper bound of S. Hence, we conclude that $m_1 = m_2$.

(b) For the sake of contradiction, we assume that $x > 0$. Let $\epsilon = x/2$ which is positive. Hence, $x \leq \frac{x}{2}$ which implies that $x \leq 0$. Contradiction! Thus, $x \leq 0$. Since $x \geq 0$, we conclude that $x = 0$.

4. In this problem, we aim to show that $\sqrt{3}+\sqrt{2}$ is irrational. We proceed as follows.
   (a) (4 points) Show that if $\sqrt{3}+\sqrt{2}$ is rational, then $\sqrt{3}-\sqrt{2}$ is also rational. *(Hint: consider their product)*
   (b) (4 points) Prove that $\sqrt{3}+\sqrt{2}$ and $\sqrt{3}-\sqrt{2}$ cannot both be rational.
   (c) (2 points) Prove that $\sqrt{3}+\sqrt{2}$ is irrational. You may use parts (a) and (b).

Solution: (a) If $\sqrt{3}+\sqrt{2}$ is rational, then $\sqrt{3}+\sqrt{2}=\frac{p}{q}$ for some $p, q \in \mathbb{Z}$ and $q \neq 0$. It is clear that $p \neq 0$, since it would imply $\sqrt{3}+\sqrt{2}=0$ otherwise. We compute $(\sqrt{3}+\sqrt{2})(\sqrt{3}-\sqrt{2})=3-2=1$, from where

$$\sqrt{3}-\sqrt{2}=\frac{1}{\sqrt{3}+\sqrt{2}}=\frac{1}{\frac{p}{q}}=\frac{q}{p}.$$

Thus, $\sqrt{3}-\sqrt{2}$ is rational.

(b) For the sake of contradiction, we assume that they are both rational. Then $(\sqrt{3}+\sqrt{2})-(\sqrt{3}-\sqrt{2})=2\sqrt{2}$ is also rational since the subtraction of two rationals is rational. $\sqrt{2}=\frac{1}{2}\cdot(2\sqrt{2})$ is rational since one half of a rational is rational. Contradiction!

(c) For the sake of contradiction, we assume that $\sqrt{3}+\sqrt{2}$ is rational. By (a), we deduce that $\sqrt{3}-\sqrt{2}$ is also rational. Thus, $\sqrt{3}+\sqrt{2}$ and $\sqrt{3}-\sqrt{2}$ are both rationals, contradicting with part (b). Thus, $\sqrt{3}+\sqrt{2}$ is irrational.